

The Vinyl Chloride Process: Safety and Environment Hazards Analysis

Mini-Project No.6
Che-396: Senior Design I
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Project Charter

Title	Goal	In Scope	Out of Scope
Senior Design Assignment 6	Investigate safety and environmental hazards of process	- HAZOP analysis - Process Hazards - Chemical Biographies	- Rest of schematic - Chemical Properties
Deliverables - HAZOP analysis for both T-100 and R-100 - Outline for Process Hazards for both T-100 and R-100 - Design Features to limit emissions for both T-100 and R-100 - Chemical Biographies		Team Members Advith Vashist Diego Martinez Tariq Hannoun Erick Yspango Jay Rawal	

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Design Plan

Task ID	Task	Start Date	Durations	Finish Date	Members
1.1	Project Charter	12/5/19	1 day	12/6/19	Advith
1.2	Design Plan	12/5/19	1 day	12/6/19	all
1.3	Objective	12/7/19	1 day		all
2.1	Ethylene	12/5/19	1 day	12/5/19	Diego
2.2	Chlorine	12/5/19	1 day	12/5/19	Diego
2.3	1,2-dichloroethane	12/6/19	1 day	12/6/19	Diego
2.4	Vinyl Chloride	12/6/19	1day	12/6/19	Diego
2.5	Hydrogen Chloride	12/7/19	1 day	12/7/19	Diego
3.1	Environmental Issues of R-100	12/7/19	1 day	12/8/19	Jay
3.2	Design Features of R-100	12/6/19	1 day	12/7/19	Jay
3.3	Process Hazards of R-100	12/6/19	1 day	12/7/19	Eric
3.4	Hazop Analysis of R-100	12/7/19	1 day	12/8/19	Eric

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4.1	Environmental Issues of T-100	12/7/19	1 day	12/8/19	Tariq
4.2	Design Features of T-100	12/6/19	1 day	12/7/19	Tariq
4.3	Process Hazards of T-100	12/6/19	1 day	12/7/19	Advith
4.4	Hazop Analysis of T-100	12/7/19	1 day	12/8/19	Advith

1. Introduction

1.1 Objective

The production of vinyl chloride involves many dangerous and hazardous chemicals. This does not mean it cannot be done but more protective measures should be taken. All chemicals involved will be analyzed before the process. Individual chemicals have different properties that make it unique from each other. The process will be analyzed to find safety and health issues. Ideas will also be given to correct any issues found. Lacking an understanding of process safety and health will lead to financial losses and worst of all, death.

Commented [1]: Feel free to add more or correct any thing.

1.2 Chemical Biography

1.2.1 Ethylene

Ethylene is a colorless gas, less dense than air, with a sweet smell and taste.[6] It is used as a raw material in the production of vinyl chloride.

Table 1 Properties of Ethylene

Chemical Formula	C ₂ H ₄
Molecular Weight	28.05
Boiling point at 1 atm	-103.70°C
Freezing Point	-169.10 °C
Heat of Combustion	-1411.20 kJ/mol
Water Solubility	Data Not Available
Flash point	-136.11 °C
Auto Ignition Temperature	450 °C
Flammability Range	25.85 %
TLV	Simple Asphyxiant (TLV-TWA)
IDLH	Data Not Available
Corrosive	No
Stability	Stable

Ethylene gas is highly flammable, flashbacks and will travel to its source of origin when ignited. At ambient temperature and pressure, ethylene rapidly or completely vaporizes.

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Vessels/containers will burst, rocket, and/or explode when heated or set on fire for a long time. When the chemical is exposed to high temperatures and pressures, it undergoes violent chemical changes. Health hazards include moderate concentration and over exposure. A moderate concentration in the air causes drowsiness, unconsciousness, and dizziness. Headache, muscular weakness, and drowsiness happen when there has been an overexposure to ethylene. Olefiant gas also causes temporary residual injury or incapacitations. Ethylene is a peroxidizable monomer that could initiate an exothermic polymerization reaction when stored in a bulk. In the presence of aluminum chloride, ethylene could react violently. Ethylene belongs to the hydrocarbons aliphatic unsaturated and polymerizable compounds groups. The following absorbents should not be used, dirt/earth, mineral-based, and clay-based absorbents. Liquids in the hydrocarbons aliphatic unsaturated and polymerizable compounds groups have been known to react with the listed absorbents.

1.2.2 - Chlorine

Chlorine is the other raw material used in the production of vinyl chloride.[4] It is a greenish yellowish gas with a strong suffocating order.

Table 2. Properties of Chlorine

Chemical Formula	Cl ₂
Molecular Weight	70.91
Boiling point at 1 atm	-34.61 °C
Freezing Point	-101 °C
Heat of Combustion	Not Pertinent
Water Solubility	0.7%
Flash point	Data Not Available
Auto Ignition Temperature	Not Flammable
Flammability Range	N/A
TLV	0.5 ppm (TLV-TWA) 1 ppm (TLV-Ceiling)
IDLH	10 ppm
Corrosive	Yes

Stability	Stable
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Chlorine is heavier than air and will settle in low areas. Under typical conditions, chlorine does not burn but like oxygen supports combustion. Active carbons may cause chlorine to ignite or explode. Chlorine belongs to oxidizing agents (strong) and halogenating agents groups. Combustible materials could ignite in its presence and mixture with fuels could cause an explosion. Steel can be ignited in the presence of rust, carbon, soot, or other catalysts at a temperature of 100 °C. Can cause an explosion at any state when it comes into contact with molten aluminum and 1-chloro-2-propyne. Reacts with copper, tin, aluminum powder, and carbides of iron. Chlorine release could be immediate or take hours and its presence would last hours with an odor threshold of 3.5 ppm. It is poisonous and if inhaled, it could cause death or serious respiratory injuries. Skin and eyes will burn if come into contact. It has 10 ppm for the IDLHs (Immediately Dangerous to Life and Health limits) rating and the following Acute Exposure Guideline Levels (AEGLs).

Table 3. AEGLs for Chlorine

Exposure Period	AEGL - 1	AEGL - 2	AEGL - 3
10 min	0.5 ppm	2.8 ppm	50 ppm
30 min	0.5 ppm	2.8 ppm	28 ppm
60 min	0.5 ppm	2 ppm	20 ppm
4 hr	0.5 ppm	1 ppm	10 ppm
8 hr	0.5 ppm	0.71 ppm	7.1 ppm

AEGLs -1 is non-disabling, transient, and reversible effects. AEGLs - 2 is irreversible or long-lasting adverse effects or impaired ability to escape. AEGLs - is life-threatening effects or death.

1.2.2 - Ethylene Dichloride

Ethylene dichloride is the product that comes out of the reactor and later broken up to make vinyl chloride.[5] It is a clear colorless liquid and has a chloroform-like scent.

Table 4 Properties of Ethylene dichloride

Chemical Formula	ClCH ₂ CH ₂ Cl
Molecular Weight	98.9

Boiling point at 1 atm	83.5 °C
Freezing Point	-35.7 °C
Heat of Combustion	3400 btu/lb
Water Solubility	no
Flash point	13.33 °C
Auto Ignition Temperature	412.78 °C
Flammability Range	9.7 %
TLV	10 ppm (TLV-TWA)
IDLH	50 ppm
Corrosive	no
Stability	Stable

At ambient temperatures, it can be ignited easily and therefore it is highly flammable. ethylene dichloride gas is heavier than air and when ignited will flashback and travel to its leaking source, even at large distances. Water will not extinguish the flames effectively. Foams, dry chemicals, and carbon dioxide should be used to extinguish any flames. Toxic and irritating gases make up the products of combustion. Inhaling ethylene dichloride vapors could cause depression, drunkenness, and nausea. Corneal injury and skin burns happen when ethylene dichloride liquid contact is prolonged. If swallowed it causes induced vomiting which can be harmful. When in gaseous state will irritate throat, nose, and eyes. Ethylene dichloride is a potential occupational carcinogen with an IDLH of 50 ppm. Currently the Department of Transportation (DOT) classifies it as poisonous. It is also incompatible with many alkali metals, nitrides, amines, azo/dizo compounds, aluminum, and epoxides. If ethylene dichloride is a halogenated aliphatic compound it will be highly reactive. When spilled in water, it could harm aquatic life and be dangerous to water intakes. If corrective response is needed, do not burn but collect and stop discharge.

1.2.3 Vinyl Chloride

VC (Vinyl chloride) is a colorless gas with a sweet scent. [8] In this process, vinyl chloride is obtained by thermal cracking of ethylene dichloride.

Table 5 Properties of Vinyl Chloride

Chemical Formula	CH ₂ CHCl
Molecular Weight	62.50
Boiling point at 1 atm	13.8 °C
Freezing Point	-153.50 °C
Heat of Combustion	-189,100 kJ/kg
Water Solubility	Slightly Soluble
Flash point	-78.89 °C
Auto Ignition Temperature	472.22 °C
Flammability Range	9.7 %
TLV	5 ppm (TLV-TWA)
IDLH	Data not available
Corrosive	No
Stability	Stable

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VC is easily ignited at ambient temperatures and pressures and is highly flammable. It has a larger density than air and when ignited will travel back, regardless of the distance, to the initial source. Protective measures should be taken in case of a fire because flashbacks will happen when VC is ignited. A react between VC, air, and water will yield an explosive product called polymeric peroxides. For small fires use dry chemicals or carbon dioxide and for large fires stop gas flow. Cool any vessel and container with VC during fires. A vessel/container will explode, rocket, or rupture if exposed to a fire, or intense heat. Highly toxic combustion products which include carbon monoxide, hydrogen chloride, and phosgene. When reacted with hot water or steam, the reaction will produce toxic fumes. If inhaled in high concentrations will cause lung irritation, dizziness, and anesthesia. When in a liquid state, will cause frostbite on skin and phenol inhibitor may be absorbed, if large amounts evaporate. VC is also a suspected carcinogen and is asphyxiating. If exposed to light a polymerization reaction could take place.

1.2.4 Hydrogen Chloride

Hydrogen chloride is a colorless to slightly yellow gas with a strong pungent odor and fumes strongly in moist air.[7] It is a byproduct of the process and it was also made in the thermal cracking of ethylene dichloride.

Table 6 Properties of Hydrogen Chloride

Chemical Formula	ClCH ₂ CH ₂ Cl
Molecular Weight	36.46
Boiling point at 1 atm	-85.0 °C
Freezing Point	-115.0 °C
Heat of Combustion	Not Pertinent
Water Solubility	82.3 % at 0 °C
Flash point	Not Flammable
Auto Ignition Temperature	Not Flammable
Flammability Range	Not Flammable
TLV	5 ppm (TLV-Ceiling)
IDLH	50 ppm

Corrosive	Yes
Stability	Stable

Hydrogen chloride is nonflammable but any containers/vessels may explode in a prolonged heat or fire. The following products of decomposition, hydrogen and chlorine, will be produced at high temperatures. Forms of stress corrosion cracking is common when hydrogen and chlorine form in high temperatures. This will cause the metal to become weak and fail, resulting in a serious accident in the future.[9] Hydrogen chloride will react violently with base metals producing hydrogen gas, which is flammable. Water is reactive with hydrogen chloride but is soluble in water. Hydrogen chloride gas is heavier than air and settles in low areas. Inhaling low or high concentrations will cause adverse side-effects regardless if short or long term exposure. Exposure of 50 to 100 ppm of hydrogen chloride gas is tolerable for about 1 hour. Brief or prolonged exposure to concentrations ranging from 1,000 ppm to 2,000 ppm is dangerous. At higher concentration, more serious respiratory distress and even death will happen. Liquid hydrogen chloride will cause burns when come on contact with skin. People with exciting respiratory and digestive problems will be severely affected by low concentrations of liquid or gas hydrogen chloride. High concentrations in water will be dangerous to aquatic life and if it enters water intakes. If spilled in water, notify health, wildlife officials, and intake operators.

1.3 Reactivity Predictions

	Ethylene			
Chlorine	Incompatible	Chlorine		
Ethylene Dichloride	Caution	Incompatible	Ethylene Dichloride	
Vinyl Chloride	Caution	Incompatible	Caution	Vinyl Chloride
Hydrogen Chloride	Incompatible	Incompatible	Caution	Incompatible

Figure 1 Predicted Reactivity Chart [3]

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The red incompatible box means, reactions will more likely form hazardous products and reaction mechanisms. Yellow caution boxes mean that hazardous reactions and products can happen under certain conditions.

2 - Process

2.1 Reactor - 100

2.1.1 R-100 for Ethylene

For the reaction of ethylene entering, the reaction is exothermic. Reaction is exothermic therefore, temperature increases and pressure increases. This process should not be vented to the air. The process has high energy levels so the process must be controlled. The chemical itself proposes moderate concentration in air causing drowsiness, dizziness, and unconsciousness. Overexposure causes headaches, drowsiness, muscular weakness. High levels of ethylene entering the air will cause wildlife to die, and damage plant life.

The current discharges is ethylene entering the air. The air should enter a membrane and should be filtered before exiting the facility. There can be a material substitution, for example the pipe being used can absorb most of the ethylene and be cleaned out when the process is complete. The last process for the discharge can be reusing the discharge ethylene for a rerun of the process or burning the ethylene off.

2.1.2 R-100 for Chlorine

Once again for the reaction, chlorine is entering and the reaction is exothermic. Reaction is exothermic therefore, temperature increases and pressure increases. This process should not be vented to the air. The process has high energy levels so the process must be controlled. After immediate exposure to a dangerous concentration of chlorine, you may experience burning pain and redness in the eye and skin. This may lead to blurred vision, and slow chest tightness. Chlorine in the air is not flammable, although, it may react to and cause explosions if reacted with other chemicals.

To properly discharge chlorine we must follow the same steps as Ethylene. Chlorine may be released into the air if no other chemical are sure not to react with the chlorine molecule. Chlorine can be mixed in with water as if a pool and then discharged. This process must not be entered into any lakes or where sea life is present. Therefore, the water must overtake the concentration of the chlorine and then the discharge can be disposed properly.

2.1.2 R-100 Process Hazards

In terms of materials, in the reactor there is a mixture of ethylene and chlorine. This mixture is incompatible as shown in figure 1.3. As previously stated in section 1.2, ethylene is a colorless gas, less dense than air, with a sweet smell and taste and is highly flammable. Ethylene enters the reactor at room temperature of 25C and exits at a temperature of 90 C. Pressure is constant as it enters as well as it exists at 1.5 atm.

The danger comes from overheating or over pressurizing inside the reactor. Under these conditions ethylene goes under violent chemical changes that lead to fires and explosions. Temperature controllers as well as pressure release valve can aid into preventing these dangers. Minor casualties can occur from a leakage in the incoming feed stream. A moderate concentration in the air causes drowsiness, unconsciousness, and dizziness. Headache, muscular weakness, and drowsiness happen when there has been an overexposure to ethylene. This can be avoided through regular maintenance of piping and instrumentation.

The only dangers that can arise from chlorine are from poor maintenance in steel components. Steel components can be ignited in the presence of rust, carbon, soot, or other catalysts at a temperature of 100 °C. Considering our stream is coming out at 90C it can be dangerous. A simple way to solve this is to use stainless steel. The biggest danger comes from human contact. If chlorine were to be released through a leak it can be fatal. Skin and eyes will burn if come into contact. It has 10 ppm for the IDLHs (Immediately Dangerous to Life and Health limits) rating and the following Acute Exposure Guideline Levels (AEGs). This can be preventable by pressure and temperature control valve to avoid overpressurizing and high level of temperature coming out of the reactor.

2.1.2 R-100 HAZOP analysis

Table 7 T-100 HAZOP analysis

Vessel: R-100 Intention: To react ethylene and chlorine to make vinyl chloride			
Guide Word	Deviation	Cause	Consequences and Action

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Stream 1 Intention: Feed Stream of Ethylene			
No	Flow	-Pipe blockage -Valves fail -Pump fails	Reactor is not able to react with chlorine. Can lead to the accumulation of chlorine in the reactor. Fixable through maintenance of the pump.
Less	Flow	-Pipe blockage -Valves fail -Pump fails	Reactor is not able to react with chlorine. can lead to the accumulation of ethylene in the reactor, which can cause the plant to explode. Ethylene is highly dangerous at high pressures. Fixable through maintenance of the pump.
More	Flow	-Control valve is fully opened -Pump speed is increased	This would cause flooding of reactor. Install high level alarm and an emergency valve to avoid catastrophic incidents.
Stream 2 Intention: Feed stream of Chlorine			
No	Flow	-Pipe blockage -Valves fail -Pump fails	Reactor is not able to react with Ethylene. Can lead to the accumulation of chlorine in the reactor. Fixable through maintenance

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			of the pump.
Less	Flow	-Pipe blockage -Valves fail -Pump fails	Reactor is not able to react with ethylene. Can lead to the accumulation of chlorine in the reactor. Fixable through maintenance of the pump.
More	Flow	-Control valve is fully opened -Pump speed is increased	This would cause flooding of reactor. Install high level alarm and an emergency valve to avoid catastrophic incidents.

Stream 3
Intention - Outlet flow of reactor of dichloroethane

No	Flow	-Pipe blockage -Valves fail -Pump fails	Reactor can no longer convert at desired conversion. Pressure can build up and result in a leak or explosion.
Reverse	Flow	- Control valve malfunction	Dichloroethane gets fed back to reactor causing reactions to fail. Install reverse flow valve that prevents backflow into the unit.

2.2 Distillation Column T-100

2.2.1 T-100 Environmental Issues, air emissions, and water discharges

Assuming that the process is controlled in the previous steps, the T-100 distillation column should contain only 1,2 dichloroethane, Vinyl chloride(VCl) and Hydrogen chloride (HCl). All three of these substances can be dangerous to personnel and/or the environment if handled poorly. Fortunately, under normal operating conditions, this distillation column is a closed system with all streams being collected or recycled entirely, and none of the streams are let out into the surroundings. If an accident were to occur and the system leaks chemicals to the environment, that damage would be significant. In the event of a fire or explosion, Vinyl chloride and 1,2 dichloroethane produce toxic vapors that would have significant effect on the atmosphere quality. Even if there isn't a fire, these two chemicals are poisonous to life in vaporous form and can harm by simply being inhaled or contacted. In the event of a liquid leak, running groundwater could transport toxins into rivers and eventually to the ocean. All three of the chemicals found in the T-100 distillation column can severely impact marine life, and every precaution must be taken to keep them in the system where they belong.

2.2.2 T-100 Design Features

Keeping the substances inside the system can be done by control systems and close monitoring. For example, the HCl must be completely vaporized in the column to ensure it's total removal from the other components. This is carried out by temperature controllers in the bottoms. The bottoms temperature is measured and adjusted by changing the flowrate of medium pressure steam in the subsequent heat exchanger. Additionally, the temperature controller is equipped with an alarm that indicates when temperatures are too low, which could potentially allow dangerous HCl into the final parts of the system. When HCl comes into contact with equipment not designed with HCl in mind, a breach is possible. With temperature controlled at a minimum value, pressure is next to be controlled. This is done both by controlling the flow rate into the condenser at the top of the column as well as by monitoring and controlling the level of fluid in the reflux drum. The condenser can cool down the outlet stream with feedback monitoring, therefore reducing the local pressure in the system. The reflux drum's level is measured and corrected via the outlet flow rate of HCl to stream 11. The final control element required for the column is to maintain the reflux ratio which is managed by flow rate controls leaving the reflux drum on its way back to the column. Last but not least, it is worth mentioning that the building materials were chosen to be stainless steel 316 for its ability to contain all of the available chemicals at all possible temperature ranges.

2.2.2 T-100 Process Hazards

In terms of materials, in the distillation column there is a mixture of HCl, dichloroethane, and vinyl chloride. The mixture is fairly unreactive except, as shown in section 1.2, dichloroethane being possibly reactive with vinyl chloride and HCl. HCl is also an OSHA PSM highly hazardous material. The threshold quantity for HCl is 5000 pounds. The reboiler will be very hot as will the outlet at 94.6 degrees celsius. Any leakage would be at a high temperature that could harm humans. Vinyl chloride and dichloroethane are also two very flammable materials.

There is a chance for overpressurizing the distillation column in instances of blockage or anything increases rate of accumulation. This would be dangerous since distillation column depends on the pressure difference between the top and bottom of the column for operation. If the pressure is not released the worst case scenarios would be an explosion or rupture. To avoid this pressure relief valves are necessary.

Depending on the type of column, structured packaging can be an ignition source. [2] Structured packaging is used because of its high efficiency that is the result of high surface area and its low pressure drop. The issue is that residue may get caught in the packaging. This could result in a fire at elevated temperatures with exposure to atmosphere. A risk when both vinyl chloride and dichloroethane are flammable.

There is a risk for loss of containment in the distillation column in case of any leakage or explosion. The explosions can be contained within an emergency valve but the leakage can only be avoided by proper maintenance of piping and instrumentation.

2.2.4 T-100 HAZOP analysis

Table 7 T-100 HAZOP analysis

Vessel: T-100 Intention: To filter out the HCl from the vinyl chloride and dichloroethane mixture			
Guide Word	Deviation	Cause	Consequences and Action
Line 10 Intention: Inlet for the flow of dichloroethane, HCl, and Vinyl Chloride to distillation column			
Reverse	Flow	Excessive vapor flow increases pressure resulting in liquid holdup in the plate above[1]	This would cause flooding of column. Could be fixed by putting in a pressure indicator that checks for sharp changes in pressure differential. If the situation is dire,

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			there could also be a plant shutdown.
No	Flow	-Pipe blockage -Valves fail -Pump fails	No operation. Could be fixed by improving maintenance procedure, installing a low level indicator, and an emergency valve to avoid catastrophic incidents.
Less	Flow	-Pipe blockage -Valves fail -Pump fails	This would change product quality. Could be fixed by improving maintenance procedure or installing a low level indicator and an emergency valve to avoid catastrophic incidents.
More	Flow	-Control valve is fully opened -Pump speed is increased	This would cause flooding of column. Install high level alarm and an emergency valve to avoid catastrophic incidents.
Line 12 Intention: Outlet for the flow of vinyl chloride and dichloroethane			
No	Flow	-Pipe blockage -Valves fail -Pump fails	No operation. Could be fixed by improving maintenance procedure or installing a low level indicator and an emergency valve to avoid catastrophic incidents.

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Less	Flow	-Pipe blockage -Valves fail -Pump fails	This would change product quality. Could be fixed by improving maintenance procedure, installing a low level indicator and an emergency valve to avoid catastrophic incidents.
More	Flow	-Control valve is fully opened -Pump speed is increased	This would cause flooding of column. Install high level alarm and an emergency valve to avoid catastrophic incidents.

Appendix

A.1 - Ethylene

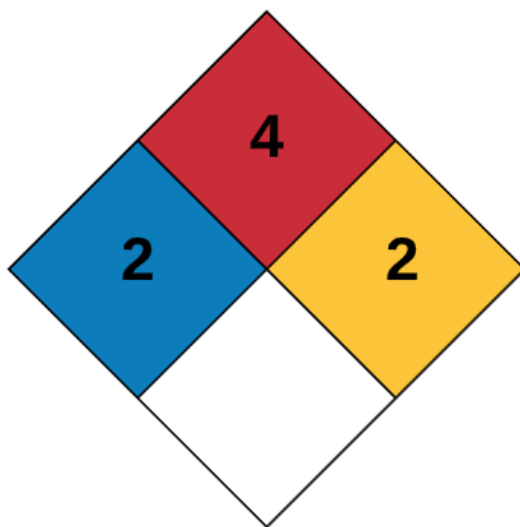


Figure 2 - NFPA Safety Square for Ethylene

Table 8 - NFPA Safety Square Meaning for Ethylene

Health (2) (Blue)	Can cause temporary incapacitation or residual injury.
Flammability (4) (Red)	Burns readily, rapidly, or completely vaporizes at atmospheric pressure and normal ambient temperature.
Instability (2) (Yellow)	Readily undergoes violent chemical changes at elevated temperatures and pressures.
Special Note (White)	None

A.2 Chlorine

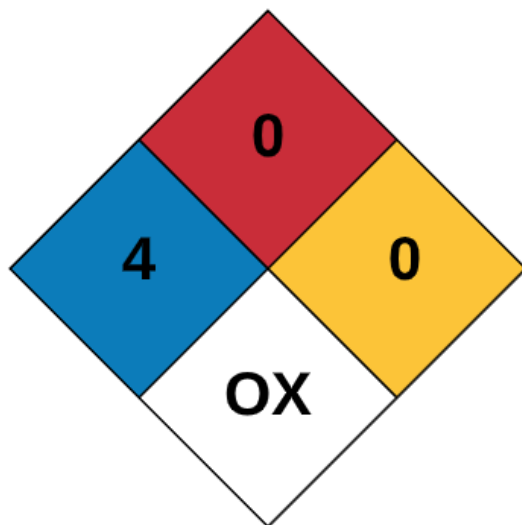


Figure 3 - NFPA Safety Square for Chlorine

Table 9 - NFPA Safety Square Meaning for Chlorine

Health (4) (Blue)	Can be lethal.
Flammability (0) (Red)	Will not burn typical fire conditions.
Instability (0) (Yellow)	Normally stable, even under fire conditions.
Special Note (OX) (White)	Possesses oxidizing properties.

Chlorine has been labeled as corrosive, poisonous, and an oxidizer by the Department of Transportation (DOT)

A.3 Ethylene Dichloride

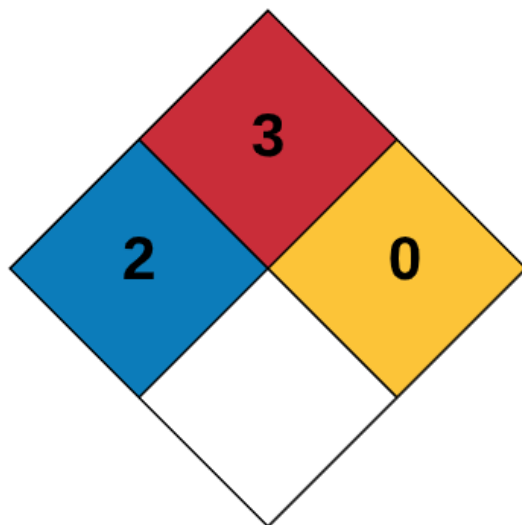


Figure 4 - NFPA Safety Square for Ethylene Dichloride

Table 10 - NFPA Safety Square Meaning for Chlorine

Health (2) (Blue)	Can cause temporary incapacitation or residual injury.
Flammability (3) (Red)	Can be ignited under almost all ambient temperature conditions.
Instability (0) (Yellow)	Normally stable, even under fire conditions.
Special Note (White)	None

A.4 Vinyl Chloride

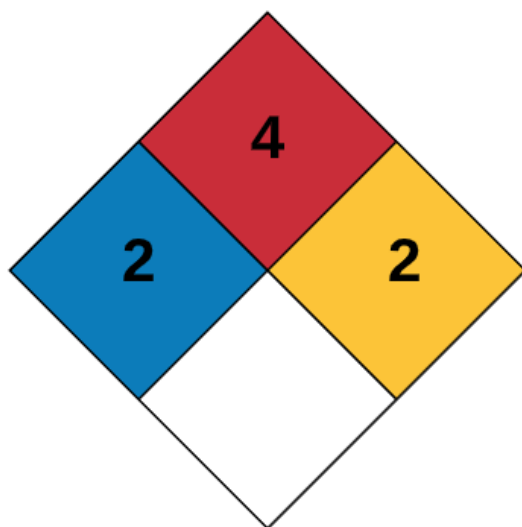


Figure 5 - NFPA Safety Square for Vinyl Chloride

Table 11 - NFPA Safety Square Meaning for Vinyl Chloride

Health (2) (Blue)	Can cause temporary incapacitation or residual injury.
Flammability (4) (Red)	Burns readily, rapidly, or completely vaporizes at atmospheric pressure and normal ambient temperature.
Instability (2) (Yellow)	Readily undergoes violent chemical changes at elevated temperatures and pressures.
Special Note (White)	None

A.5 Hydrogen Chloride

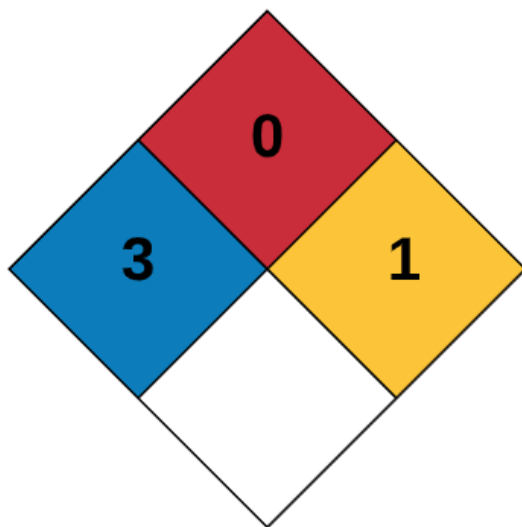


Figure 6 - NFPA Safety Square for Hydrogen Chloride

Table 12 - NFPA Safety Square Meaning for Hydrogen Chloride

Health (3) (Blue)	Can cause serious or permanent injury.
Flammability (0) (Red)	Will not burn under typical fire conditions.
Instability (1) (Yellow)	Normally stable but can become unstable at elevated temperatures and pressures.
Special Note (White)	None

DOT has labeled hydrogen chloride as corrosive and poisonous.

References

- [1] Tham, M T. *Factors Affecting Distillation Column Operation*, <https://www.rccostello.com/distil/distilop.htm>.
- [2] "Minimize the Risk of Fire During Distillation Column Maintenance." *Ntnu*, AiCHE, http://folk.ntnu.no/skoge/prost/proceedings/distillation10/DA2010 Sponsor Information/Koch Glitsch/Technical_Articles/Packing/Minimize_Risk_of_Fire.pdf.
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